

Aditya Shibu

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Dubai, UAE

SUMMARY

Undergraduate researcher in autonomous systems and neural compression with two published conference papers and a journal preprint targeting RA-L. Experienced in end-to-end robotics pipelines spanning perception, planning, and edge deployment, with hands-on work in C++, CUDA, TensorRT, and ROS 2. 1st Place winner at Formula Student AI (Overall Statics and Real World AI).

EDUCATION

- Bachelor's of Science (Hons.) - Computer Science with specialization in AI** **GPA: 4.0/4.0**
Heriot Watt University, Dubai, UAE *Sept 2023 - Present*

RESEARCH AND PUBLICATIONS

- Simulating an Autonomous System in CARLA Using ROS2** **Research Paper** - Published
International Conference on Signal Processing and Information Security (ICSPIS) *November 2025*
- SkySim: Natural Language Control of Drone Swarms using LLMs** **Research Paper** - Published
International Conference on Drones and Unmanned Systems (DAUS) *February 2026*
- LiZIP: A Deep Learning Model for Compression of LiDAR point clouds** **Research Paper** - Preprint
School of Mathematical and Computer Sciences - Heriot Watt University *Target - RA-L*

SKILLS

- Languages:** C, C++, Python, CUDA, JavaScript, Java, OCaml
- AI / Robotics:** PyTorch, TensorFlow, TensorRT, YOLO, ROS2, CARLA, Gazebo, Ollama
- Web / General:** ReactJS, NextJS, ExpressJS, TailwindCSS, MongoDB, MySQL, FireStore
- Tools:** Linux, Git, GitHub Actions, Docker

WORK EXPERIENCE

- Lab for AI and Robotics Research (LAIRR) - Heriot Watt University** **Dubai, UAE**
Research Intern *Nov 2025 - Present*
 - Engineered **LiZIP**, a novel auto-regressive neural compression architecture for LiDAR point clouds, writing the full inference pipeline from scratch in C++, custom CUDA kernels, and TensorRT.
 - Optimized for bare-metal edge deployment on the NVIDIA Jetson AGX Orin, achieving real-time inference latency of **16.7ms to 26.8ms** per frame, **outperforming LASzip by up to 2x in speed** while consuming <20% CPU and <1% GPU utilization.
 - Successfully outperformed industry standards **LASzip** and **Google Draco** across both datasets in compression efficiency, reducing payload sizes to **186 KB and 605 KB** respectively, while maintaining a near-lossless maximum error of **<0.017 mm**.
- AimbeyondD - Heriot Watt University** **Dubai, UAE**
Research Intern *Nov 2025 - Present*
 - Developed **SkySim**, a ROS 2 Jazzy-based simulation environment on Gazebo for Natural Language Control of Drone Swarms using LLMs, with **Artificial Potential Fields**-based collision avoidance and formation flying
 - Integrated **Ollama** for on-device SLM inference with **ROS 2** nodes for drone swarm communication, enabling low-latency translation of high-level commands into low-level control instructions
 - Deployed on **10x Bitcraze 2.1 Brushless Drones** using **Optitrack** motion capture for real-time position streaming to SLM
- Atlas Racing FS** **Dubai, UAE**
Head of Driverless & Lead Autonomous Systems Developer *May 2025 - Present*
 - Developed end-to-end autonomous vehicle pipeline in **CARLA** with **deep neural networks for perception** and path planning (**Pure Pursuit, MPC**) integrated with **ROS 2 Humble**, achieving seamless simulation-to-reality transfer for Formula Student AI
 - Built comprehensive perception pipeline using **ZED2i stereo cameras** and **RoboSense Helios 16 LiDAR**, implementing **DBSCAN clustering** and sensor fusion for object detection and localization
 - Optimized **YOLO inference with TensorRT acceleration**, reducing latency by **88%** (140ms → 16ms) and achieving stable **40-50 FPS** for real-time autonomous vehicle operation
 - Led technical development resulting in **1st Place Overall Statics** and **1st Place Real World AI**, demonstrating production-ready autonomous systems performance
- Center for AI and Robotics (CAIR) - New York University** **Abu Dhabi, UAE**
Research Collaborator *Dec 2025 - Feb 2026*
 - Architected and trained a **Conditional Imitation Learning (CILRS)** neural network using PyTorch to navigate complex urban intersections based on high-level GPS commands.
 - Configured and managed headless CARLA simulations on dual **NVIDIA L40s HPC** infrastructure, building synthetic urban driving datasets (RGB + Telemetry) for end-to-end driving models.